

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404769
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Salvino; Chris et al.

Heat recovery arrangement

Abstract

Described herein are embodiments directed to a heat recovery arrangement for collecting vaporized gas trapped in regolith. The heat recovery arrangement generally comprising a rover that carries heat recovery elements that cooperate with a primary heat source. The heat recovery elements include a preheat contact element that preheats a region of regolith before the region is brought to high heat by the primary heat source. As the rover moves forward, the preheat contact element receives heat collected from the high heat region via a heat recovery sled that moves in contact with the high heat region. Heat is transferred between the heat recovery sled and the preheat contact element via a heat transfer medium that circulates through the heat recovery sled and preheat contact element.

Inventors: Salvino; Chris (Scottsdale, AZ), DeJarnette; Drew (San Diego, CA), Beatty; Paul A (Fort Collins, CO)

Applicant: LUNAR HELIUM-3 MINING, LLC (Scottsdale, AZ)

Family ID: 96882050

Assignee: LUNAR HELIUM-3 MINING, LLC (Scottsdale, AZ)

Appl. No.: 19/079201

Filed: March 13, 2025

Related U.S. Application Data

continuation-in-part parent-doc US 18798615 20240808 PENDING child-doc US 19079201

Publication Classification

Int. Cl.: E21C51/00 (20060101); B64G1/16 (20060101); F24S20/00 (20180101); F24S23/70 (20180101); F24S23/71 (20180101); F24S80/20 (20180101)

U.S. Cl.:

CPC **E21C51/00** (20130101); **B64G1/16** (20130101); F24S20/00 (20180501); F24S23/71 (20180501); F24S2023/878 (20180501); F24S80/20 (20180501)

Field of Classification Search

CPC: B10D (7/02); E21C (51/00); F24S (23/71); F24S (23/12); F24S (40/55); B64G (1/16)

USPC: 126/684-685; 126/688-690; 126/568-713; 405/128.55; 405/128.6; 96/4

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3536908	12/1969	Oster	362/559	F21S 10/005
3638312	12/1971	Szwarc	433/25	G02B 6/4298
3985116	12/1975	Kapany	428/179	F24S 80/50
4026267	12/1976	Coleman	136/246	F24S 23/30
4201197	12/1979	Dismer	126/685	F24S 23/71
4240692	12/1979	Winston	136/246	G02B 6/262
4282858	12/1980	Bowers, Jr.	126/685	F24S 50/20
RE31678	12/1983	Ochiai	385/115	F21S 11/00
4511755	12/1984	Mori	136/246	F21S 11/00
5168158	12/1991	McComas et al.	N/A	N/A
7514694	12/2008	Stephan et al.	N/A	N/A
7919758	12/2010	Stephan et al.	N/A	N/A
7973235	12/2010	Muhs	136/246	F24S 23/79
8063300	12/2010	Horne	136/246	H10F 77/488
8330115	12/2011	Frank	N/A	N/A
9134047	12/2014	Black et al.	N/A	N/A
9261468	12/2015	Bingham et al.	N/A	N/A
9599729	12/2016	Roscoe et al.	N/A	N/A
10222121	12/2018	Cullinane et al.	N/A	N/A
11624542	12/2022	Sung	N/A	N/A
12091973	12/2023	Salvino	N/A	E21C 51/00
2012/0017888	12/2011	Garza	126/678	F24S 23/12
2013/0034198	12/2012	Chandrasekharan et al.	N/A	N/A
2016/0102890	12/2015	Saucedo	126/690	F24S 23/79
2017/0146202	12/2016	Stone	N/A	F21K 9/61
2019/0200510	12/2018	Chrysanthakopoulos	N/A	A01B 59/043
2019/0200519	12/2018	Chrysanthakopoulos	N/A	B25J 9/1697
2020/0240267	12/2019	Sercel	N/A	B64G 99/00
2022/0090500	12/2021	Sercel	N/A	E21C 51/00
2023/0000003	12/2022	Chrysanthakopoulos	N/A	A01M 21/046
2023/0304403	12/2022	Kuhns	N/A	B64G 1/16
2023/0411134	12/2022	Ryan et al.	N/A	N/A
2024/0035379	12/2023	Grillos	N/A	N/A

OTHER PUBLICATIONS

Primary Examiner: Shirsat; Vivek K

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This continuation-in-part application which claims priority to and the benefit of U.S. patent application Ser. No. 18/798,615 entitled: COMPOUND PARABOLIC CONCENTRATOR ARRANGEMENT, filed on Aug. 8, 2024, the entire disclosure of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates generally to heat recovery for mining extraterrestrial sites for gas trapped in extraterrestrial soil.

2. Description of Related Art

(2) Helium-3 (He-3) holds immense potential as a clean and efficient energy source, offering numerous benefits for various applications. It has been discovered that He-3 is a highly efficient fuel for nuclear fusion, which has the potential to provide a nearly limitless and environmentally friendly energy solution. When He-3 fuses with deuterium, it produces a helium atom, a proton, and a tremendous amount of energy. Unlike traditional nuclear fission, fusion reactions release energy without generating hazardous radioactive waste or greenhouse gas emissions. Harnessing He-3 for fusion could revolutionize the energy landscape, providing a stable and sustainable source of power for the future.

(3) However, despite its remarkable benefits, mining He-3 on Earth faces significant limitations. For one thing, He-3 is incredibly scarce on our planet. It is primarily found in minute quantities in the top surfaces of lunar soil, known as regolith, and is sparsely present in the Earth's atmosphere. Consequently, the extraction of He-3, whether on Earth or the Moon, is challenging and expensive. Moreover, mining it on the Moon poses logistical difficulties requiring advanced technologies and significant investments. Even assuming mining on the Moon is successful, transporting the mined He-3 from the Moon to Earth presents significant technical challenges and high costs. These limitations highlight the need for further research and development to find more accessible and cost-effective sources of He-3 or to explore alternative fusion fuel options that do not rely solely on He-3.

(4) The subject matter disclosed herein is generally directed to innovations related to collecting He-3 and other useable gaseous elements on extra-terrestrial bodies.

SUMMARY OF THE INVENTION

(5) The present invention generally relates to heat recovery to be used to improve extraction efficiency for gases trapped in extraterrestrial soil.

(6) Certain embodiments of the present invention envision a heat recovery arrangement generally comprising a rover carrying a primary heat source that cooperates with an arrangement of heat recovery elements. The rover can be defined as having wheels or tracks that move the rover over an extra-terrestrial surface. The rover can be defined between a rover front end and a rover rear end, wherein the rover is configured to travel in a direction leading with the rover front end. The heat recovery elements include a preheat contact element and a heat recovery sled, both of which are configured to rest atop granular soil. The heat recovery elements further include a gap that separates the preheat contact element from the heat recovery sled, wherein the preheat contact

element is closer to the rover front end than the heat recovery sled. The heat recovery arrangement further includes a primary heat source (configured to heat the granular soil in the gap) and a gas collecting shield covering at least the gap. The heat recovery elements further include a heat exchange tube comprising a feed line, a return line, a heat collecting coil, and a heat emitting coil, wherein the feed line is configured to flow heated medium from a heat collecting coil to the heat emitting coil, the return line is configured to flow cooled medium from the heat emitting coil to the heat collecting coil. The heat recovery sled contains the heat collecting coil and the preheat contact element contains the heat emitting coil. The gas collecting shield, the preheat contact element and the heat recovery sled are located under the rover.

(7) Still other embodiments envision a heat recovery system generally comprising a rover carrying a primary heat source that cooperates with an arrangement of heat recovery elements, the rover being configured to move in a forward direction. The heat recovery elements can include a preheat contact element that is configured to contact and preheat granular soil and a heating gap between the preheat contact element and a heat recovery sled. The heat recovery system further includes a primary heat source that is configured and arranged to heat a region of the preheated granular soil in the gap. The heat recovery sled is configured and arranged to collect at least a portion of the heat from the region when the rover moves the heat recovery sled in contact with the region. The heat recovery system further includes a gas collecting shield covering at least the gap, wherein the gas collecting shield is configured to contain gas liberated from the granular soil when the granular soil is heated by the primary heat source. The heat recovery elements further include a heat exchange tube comprising a feed line (that is configured to carry warmed medium from a heat collecting coil in the preheat contact element to a heat emitting coil in the heat recovery sled) and a return line (that is configured to carry cooled medium from the heat emitting coil to the heat collecting coil). The gas collecting shield, the preheat contact element and the heat recovery sled are located on an underside the rover.

(8) While other embodiment of the present invention contemplates a method for using one or more rovers supporting a heat recovery system wherein the method generally comprises steps for heating granular soil, such as regolith, while moving the rover. More specifically, the method can include a heating step for heating a first region of granular soil in a gap between a preheat contact element and a heat recovery sled with a primary heat source. Moving the heat recovery sled in contact with the first region of granular soil. Warming a medium that is circulating through a heat collecting coil while the heat recovery sled is in contact with the first region of granular soil. The coil being embedded in or otherwise snaking through the heat recovery sled. The warming step is accomplished by collecting a portion of the heat from the heated granular soil. The method further includes preheating a second region of granular soil by circulating the warmed medium through a heat emitting coil embedded in the preheat contact element. Moving the gap over the preheated second region of granular soil and heating the preheated second region of granular soil via the primary heat source.

(9) Yet other embodiments of the present invention contemplate a CPC heating arrangement that generally comprises a transporter that carries a plurality of CPCs that are equipped to heat granular soil to mine target gases trapped in the granular soil. More specifically, the transporter comprises a transporter body defining a top end and a bottom end, wherein the bottom end is configured to interface a surface of an extra-terrestrial body. The transporter is configured to be moved in different locations on the extra-terrestrial body, such as the moon. The CPC arrangement further comprises a cover having an internal cover volume, which is a volume within the cover defined from between a cover rim and an inside cover top of the cover. The internal cover volume defines a shielded environment when the cover rim rests atop granular soil. The shielded environment is the internal volume of the cover and is only in communication with an outside environment via the cover rim when the cover rim is not resting atop the granular soil. Resting atop means in contact with the granular soil wherein there are essentially no gaps between the granular soil surface and

the cover rim. Resting can be wherein at least a portion of the weight of the cover is supported by the granular soil or if the cover is pressed into the granular soil via a motorized actuator connected to the transporter. The plurality of CPCs extends from the top end, such as an array on an armature or frame. Each CPC comprises a concave reflective bowl having a refractive index of at least 1.4. Each of the concave reflective bowls are defined between a bowl rim and a bowl apex, wherein the bowl rim is configured to confront the sun and the bowl apex is configured to receive sunlight from the concave reflective bowl. Each of the concave reflective bowls further comprises a fiberoptic cable extending from a fiberoptic inlet end connected to the bowl apex to a fiberoptic outlet end located in the internal volume. The fiberoptic outlet end is configured to reside within 6 inches from the granular soil. The sunlight is configured to be received at the fiberoptic inlet end and emitted at the fiberoptic outlet end.

(10) Another embodiment of the present invention contemplates a CPC arrangement generally comprising a transporter, an internal cover volume and a cryogenically cooled surface. The transporter supports a plurality of CPCs each configured to receive sunlight. Each CPC comprises a concave reflective bowl defined between a bowl rim and a bowl apex comprising a bowl aperture and a fiberoptic cable extending from a fiberoptic inlet end at the apex to a fiberoptic outlet end. The fiberoptic outlet end is configured to emit the sunlight received at the fiberoptic inlet end from the bowl aperture on a granular surface. The CPC arrangement further comprises an internal cover volume defined within a cover that extends from an internal cover top to a cover rim. There is a cryogenically cooled surface within the internal cover volume, wherein the cryogenically cooled surface is configured to be cooled by cryogenic liquid. The fiberoptic outlet end extends into the internal cover volume.

(11) Still another embodiment of the present invention contemplates a CPC system that generally comprises a transporter that carries a plurality of CPCs that are equipped to heat regolith to mine target gases trapped in the regolith. More specifically, the transporter defines a top end and a bottom end with the bottom end being configured to interface regolith. The transporter configured to be moved in different locations on the moon. The CPC system further comprises a cover that defines a shielded environment when a cover rim of the cover rests atop the regolith. The shielded environment (or the internal cover volume when not in contact with the regolith) is only in communication with an outside environment via the cover rim when the cover does not rest atop the regolith. The plurality of CPCs extends from the top end, wherein each CPC comprises a concave parabolic reflective bowl. Each of the concave reflective bowls is defined between a bowl rim and a bowl apex, wherein the bowl rim is configured to receive sunlight. An aperture in the bowl apex is configured to receive the sunlight from the concave reflective bowl. A fiberoptic cable extends from a fiberoptic inlet end to a fiberoptic outlet end and connects to the bowl apex at the fiberoptic inlet end. The fiberoptic cable is in light communication with the aperture, meaning light moves through the aperture and into the fiberoptic cable. The fiberoptic outlet end is in the shielded environment and is configured to interface the regolith. The fiberoptic cable is configured to transmit the sunlight from the fiberoptic inlet end through the fiberoptic outlet end.

(12) Certain other embodiments of the present invention envision a CPC and gas collection arrangement that generally comprises a transporter that carries a plurality of CPCs that are equipped to heat granular soil to mine target gases trapped in the granular soil. More specifically, the transporter is configured to be moved in different locations on the moon. The arrangement further comprises a cover that defines a shielded environment when a cover rim of the cover rests atop the regolith. The shielded environment is not in communication with an outside environment. Each of the plurality of CPCs comprises a concave parabolic reflective bowl defined between a bowl rim and a bowl apex. The bowl rim is configured to receive sunlight and direct the sunlight to the aperture at the bowl apex.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-1D are line drawings of a compound parabolic concentrator and gas collection arrangement constructed in accordance with embodiments of the present invention;
- (2) FIGS. 2A and 2B are line drawings of different perspectives of a rover pulling a wagon carrying a CPC and gas collection arrangement in accordance with embodiments of the present invention;
- (3) FIG. 3A is a line drawing of the CPC and gas collection arrangement of FIG. 1A between the cross-section lines AA and BB (of FIG. 2B) constructed in accordance with embodiments of the present invention;
- (4) FIG. 3B is a line drawing illustratively depicting a cross section of the CPC and gas collection arrangement of FIG. 1C constructed in accordance with embodiments of the present invention;
- (5) FIGS. 3C and 3D are line drawings illustratively depicting cross-sections of an embodiment of a motorized convex CPC arrangement in two different positions on a wagon consistent with embodiments of the present invention;
- (6) FIG. 3E is a line drawing illustratively depicting a side view of another CPC and gas collection arrangement with a convex array doubling as a cover consistent with embodiments of the present invention
- (7) FIG. 4A, in view of FIG. 4B, is a line drawing of an isometric view of a CPC and fiberoptic cable arrangement consistent with embodiments of the present invention;
- (8) FIG. 4B is a line drawing of a CPC cross-section depicting solar energy concentration consistent with embodiments of the present invention;
- (9) FIG. 4C is a line drawing of a compound parabolic concentrator array cross-section in a convex arrangement consistent with embodiments of the present invention;
- (10) FIGS. 5A and 5B are line drawings illustratively depicting a redirecting spring and nitinol deflector consistent with embodiments of the present invention;
- (11) FIGS. 6A-6G are line drawings illustratively depicting a heat recovery arrangement that can be used in conjunction with the CPC and gas collection arrangement consistent with embodiments of the present invention; and
- (12) FIG. 6H is an active circulation system connected to the heat exchange tubes consistent with embodiments of the present invention.

DETAILED DESCRIPTION

(13) Initially, this disclosure is by way of example only, not by limitation. Thus, although the instrumentalities described herein are for the convenience of explanation, shown and described with respect to exemplary embodiments, it will be appreciated that the principles herein may be applied equally in other similar configurations involving the subject matter directed to the field of the invention. The phrases “in one embodiment”, “according to one embodiment”, and the like, generally mean the particular feature, structure, or characteristic following the phrase, is included in at least one embodiment of the present invention and may be included in more than one embodiment of the present invention. Importantly, such phrases do not necessarily refer to the same embodiment. If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic. As used herein, the terms “having”, “have”, “including” and “include” are considered open language and are synonymous with the term “comprising”. Furthermore, as used herein, the term “essentially” is meant to stress that a characteristic of something is to be interpreted within acceptable tolerance margins known to those skilled in the art in keeping with typical normal world tolerance, which is analogous with “more or less.” For example, essentially flat, essentially straight, essentially on time, etc. all indicate that these characteristics are not capable of being perfect within the sense of their limits. Accordingly, if there

is no specific+/-value assigned to “essentially”, then assume essentially means to be within +/-2.5% of exact. The term “connected to” as used herein is to be interpreted as a first element physically linked or attached to a second element and not as a “means for attaching” as in a “means plus function”. In fact, unless a term expressly uses “means for” followed by the gerund form of a verb, that term shall not be interpreted under 35 U.S.C. § 112 (f). In what follows, similar or identical structures may be identified using identical callouts.

(14) With respect to the drawings, it is noted that the figures are not necessarily drawn to scale and are diagrammatic in nature to illustrate features of interest. Descriptive terminology such as, for example, upper/lower, top/bottom, horizontal/vertical, left/right and the like, may be adopted with respect to the various views or conventions provided in the figures as generally understood by an onlooker for purposes of enhancing the reader's understanding and is in no way intended to be limiting. All embodiments described herein are submitted to be operational irrespective of any overall physical orientation unless specifically described otherwise, such as elements that rely on gravity to operate, for example.

(15) Described herein are embodiments directed to collecting gaseous atoms and molecules in an extremely low-pressure environment and collecting those gaseous atoms and molecules using cryogenically cooled surfaces, such as plates. Extremely low-pressure environment is defined herein as below 1×10^{-5} bars, wherein pressure at sea-level on Earth is approximately 1 bar. Aspects of the present invention consider mining gaseous atoms and molecules on extraterrestrial bodies such as the Moon, asteroids, moons orbiting other planets, etc., for example. Many of these extraterrestrial bodies have little to no ambient pressure at or just beyond their surfaces and depending on the size of the extraterrestrial body have a significantly lower gravitational pull than that experienced on Earth.

(16) Though embodiments of the present invention can be used in conjunction with many different extraterrestrial bodies, it is one object of the present invention to focus on mining gaseous atoms and molecules (materials) from the Moon. The Moon's atmosphere (the surface boundary exosphere) has a pressure of about 3×10^{-15} bars and can range in temperature between 20° to 400° Kelvin. In the interest of continued exploration of the Moon as well as maintaining long-term residents on the Moon, extracting or otherwise mining important gaseous materials from the Moon, such as oxygen, nitrogen, hydrogen, and helium, for example, reduces the dependency of transporting such gaseous materials from the Earth. Moreover, helium-3 (He-3), a light stable isotope of helium having two protons and one neutron, which has promise as an important constituent in fusion reactions, is far more abundant on the Moon than the Earth. In some estimates, helium-3 is more than a thousand times more abundant on the Moon than compared to the Earth making the Moon a better target to obtain helium-3.

(17) Certain embodiments of the present invention envision extracting target gaseous materials from the Moon by heating up moon regolith (lunar soil/minerals) to desorb valuable gas, such as He-3, that is bound to the regolith. A target gas is a specific gas that is desired to be obtained. Desorbing temperatures are defined as temperatures that are high enough to liberate/desorb (and possibly vaporize) these target gaseous materials from moon regolith, or simply “regolith”. Desorbing temperatures may start around 600° C. The desorbed target gaseous materials such as He-3 are then collected as liquid from condensation surfaces that are at or below the condensation temperatures corresponding to each of the target gaseous materials. When condensed i.e., liquefied, the liquid or frozen, which improves the transportation of these target materials.

(18) Presented below are embodiments directed to collecting gas via a CPC arrangement. The CPC arrangement generally comprises a transporter that carries around a cover with one or more cryogenically cooled condensation plates (or some other cryogenically cooled condensation surface) therein. A plurality of CPCs dispersed on the transporter each have a concave reflective bowl that captures and directs sunlight through a fiberoptic cable where the sunlight is focused on regolith in an internal environment defined within the cover when resting atop the regolith. The

focused sunlight heats the regolith and liberates the gas from the regolith, which is trapped in the internal environment. The gas in the internal environment condenses on the cooled plates where it can be collected and processed.

(19) Referring to the drawings in general, FIGS. **1A-1D** are line drawings of a compound parabolic concentrator and gas collection arrangement constructed in accordance with embodiments of the present invention. FIG. **1A** is an isometric view of a compound parabolic concentrator and gas collection arrangement **100** depicting a full semi-spherical convex compound parabolic concentrator (CPC) array **122** disposed on a rover wagon **102b**. Certain embodiments envision photo voltaic solar (PV) panels **216** being disposed in the spaces, or gaps, between the CPCs **120** to passively generate electricity for the transporter **102a**. Though there are only three PV panels **216** shown here, there can be enough PV panels **216** to fit between all the outer bowl rims **134** of the CPCs **120**. Each CPC **120** comprises a bowl aperture **139** at the bowl apex **136** in the bottom of the CPC bowl **130**. The bowl apertures **139** are not depicted in the other figures for sake of simplicity but are nonetheless an integral part of the CPC **120**. FIG. **1B** is an isometric view of an optional CPC and gas collection arrangement **150** depicting a half semi-spherical convex CPC array **122b** disposed on a rover wagon **102b**. This further shows the fiberoptic cables **124** extending from the CPCs **120** into the wagon body **104** of the rover wagon **102b**. FIG. **1C** is an isometric view of yet another optional CPC and gas collection arrangement **160** depicting a full semi-spherical concave CPC array **162** disposed on a rover wagon **102b**. FIG. **1D** is an isometric view of still another optional CPC and gas collection arrangement **170** depicting a half semi-spherical concave CPC array **162b** disposed on a rover wagon **102b** with the fiberoptic cables **124** extending from the CPCs **120** into the body of the rover wagon **102b**. The concepts of the embodiments of FIGS. **1A-1D** are described with respect to forgoing embodiment.

(20) FIGS. **2A** and **2B** are line drawings of different perspectives of a rover **102** pulling a wagon **102b** carrying a CPC and gas collection arrangement **100** in accordance with embodiments of the present invention. Though the wagon **102a** is the transporter **102a** supporting or otherwise carrying the CPC and gas collection arrangement **100**, the transporter **102a** can be a rover **102**, a robot or human carried platform or some other support structure that is arranged to support CPCs **120** in a manner that accomplishes the tasks described herein.

(21) FIG. **2A** is a side view of the CPC and gas collection arrangement **100**, depicting the rover **102** connected or otherwise linked to the wagon **102b** via a hitch linkage **105**. The CPC and gas collection arrangement **100** is depicted with the semi-spherical convex CPC array **122** mounted on the rover wagon **102b**. Both the rover wagon **102b** and the rover **102** comprise a cover rim **111** configured to essentially press (or somewhat seal) against the granular surface **112a** (which when referring to the moon **141**, is regolith surface) to form the shielded environment **114** inside of the cover **110**. Some embodiments envision the rover **102** not comprising a shielded environment but rather used to simply pull one or more wagons **102b** around the surface of the moon **141** or some other extra-terrestrial body. The shielded environment **114** is essentially the same volume as the internal volume of the cover **110** defined from the inside cover top **107c** to the cover rim **111**, which by convention will also have the callout **114**.

(22) With respect to the rover **102**, general components can include supporting elements that provide all necessary power and functionality needed to operate the rover **102** and the wagon **102b**. This includes cryogenic fluid tanks and pumping system **202**, a battery system **204**, communication system **210**, computer system **205**, etc. The cryogenic fluid tanks and pumping system **202** (as viewed in conjunction with FIG. **3A**) can provide a cooling liquid to cool condensation plates **115** in the gas collection system **118**. A cryogenic liquid pumping station **117**, that can be connected to the cryogenic fluid tanks **202** via fluid carrying lines (not shown), circulates cryogenic fluid through conduits **119** either encased in/or lining the gas condensation plates **115**. Gaseous atoms and molecules that are liberated from the regolith **112** (via heat, for example) condense on the cryogenically cooled gas condensing plates **115** inside of the shielded environment **114**.

(23) With continued attention to FIG. 2A, the rover **102** can comprise a battery system **204** that can include one or more batteries that can be rechargeable by photo-voltaic panels or some internal generator that uses fuel, or some combination thereof. The communications system **210** can be a wireless communicator that uses RF, laser, IR or some other wireless communications means understood by those skilled in the art. The computer system **205** generally includes a central processing unit, processors, volatile/transient and non-transient memory, etc., and can function within a computer area network contained within the rover **102** and CPC and gas collection arrangement **100**. Electric driving motors **212** can be built into the wheel hubs to spin the wheels **208** (or tracks) that propel the rover **102** around the surface **112a**. The suspension system **206** provides flexibility to the drivetrain to navigate over uneven terrain for the rover **102**, and wagon **102b**.

(24) FIG. 2B is a top view line drawing of the rover **102** pulling the wagon **102b** carrying the CPC and gas collection arrangement **100**. The rover **102** presents the top surface **107** of the rover **102** calling out the cryogenic liquid tanks **202**, batteries **204** and the communication system **210**. The CPC and gas collection arrangement **100** is sectioned by two cross-section lines AA and BB through a plurality of CPCs **120** along the center of the wagon **102b**.

(25) FIG. 3A is a line drawing of the CPC and gas collection arrangement of FIG. 1A between the cross-section lines AA and BB (of FIG. 2B) constructed in accordance with embodiments of the present invention. As depicted, the CPC and gas collection arrangement **100** generally includes a transporter that supports a convex CPC array **122**, which is configured to concentrate and direct solar energy as heat to granular soil **112** within the footprint **111a** of a shielded environment **110**. The dashed lines depict components of interest that are not between the cross-section lines AA and BB.

(26) As viewed with respect to FIGS. 4B and 4C, the convex CPC array **122** comprises a plurality of CPCs **120** distributed in an arched/non-planar arrangement. The CPCs **120** can be disposed along a semi-spherical frame **121** that is an armature that supports the plurality of CPCs **120**. Each CPC **120** is configured to capture solar energy **142** and direct the solar energy in a concentrated form **143a** into an inlet end **128** of a fiberoptic light cable **124** (also known as a light pipe). The concentrated solar energy **143a** is transferred through the fiberoptic cable **124** and emitted from the outlet end **126** of the fiberoptic cable **124** as a heat beam that is directed at the regolith **112**. The outlet end **126** is disposed within the internal volume/shielded environment **114**. The shielded environment **114** is defined by and within the cover **110** when resting atop the regolith **112**, which happens to be essentially the same volume as the internal volume of the cover from the inside cover top **107c** to the cover rim **111**. In this embodiment, the cover **110** extends (via cover sides **101**, in this configuration) from the inside of a cover top **107c** towards the regolith surface **112a**. In certain embodiments, the cover **110** can include a pliable or moveable skirt **113**, which can actuate towards or away from the regolith surface **112a**, as shown by the arrow **119**. Here, the skirt **113** is at least somewhat retracted to allow clearance for the wagon **102b** to roll over the regolith **112** (as shown by the wheels **108** partially revealed). In this embodiment the distal rim of the moveable skirt **113** serves as the cover rim **111**. The cover rim **111** is configured to seal against the regolith surface **112a**, thereby creating the shielded environment **114** within the cover **110**. The cover rim **111** defines the footprint **111a** of the shielded environment **114**. The shielded environment **114** (internal volume) is only in communication with the outside environment **116** when the cover rim **111** is not in contact with the regolith surface **112a**. Accordingly, a high heat light concentrated beam **143b** from the concentrated solar energy **143a** is emitted from each fiberoptic outlet ends **126** and directed to regolith **112**. Gaseous elements trapped in the regolith **112** are liberated when temperatures exceed their liberation threshold, which for some elements and molecules is between 600° C.-1000° C. The liberated gaseous elements contained in the internal environment **114** can then be collected by gas collectors **115** inside of or at least in fluid communication with the internal environment **114** (i.e., in a separate container having an environment that can receive the gases

from the internal environment **114**).

(27) In this embodiment, the wagon **102b** supports the cryogenic liquid pumping station **117** and batteries **204**, which are bolted or otherwise attached to the wagon's top surface **107b**. The cryogenic liquid pumping station **117** circulates cryogenic fluid (received from the cryogenic liquid tanks **202** on the rover **102**) through the liquid cryogen carrying conduits **119**, which in this embodiment run through the gas condensation plates **115**.

(28) FIG. 3B is a line drawing illustratively depicting a cross section of the CPC and gas collection arrangement **160** of FIG. 1C constructed in accordance with embodiments of the present invention. As shown, the concave curved CPC array **162** is arched in a concave arrangement. Here, the CPCs **120** are disposed along a concave semi-spherical frame **161**, which is adapted and arranged as an armature to support the plurality of CPCs **120**. The concave curved CPC array **162** can be static or optionally dynamically moved to follow the sun **140** as depicted in FIGS. 2C and 2D.

(29) FIGS. 3C and 3D are line drawings illustratively depicting cross-sections of an embodiment of a motorized convex CPC arrangement **165** in two different positions on a wagon **102b** consistent with embodiments of the present invention. As shown in FIG. 3C the concave curved CPC array **162** is angled to more efficiently collect solar energy from the sun **140** when the sun **140** is behind the wagon **102b**, or when the sun **140** is in front of the wagon **102b**, as shown in FIG. 2D. The concave curved CPC array **162** can be moved by a motor (not shown) from the direction depicted in FIG. 3C to that of FIG. 3D.

(30) FIG. 3E is a line drawing illustratively depicting a side view of another CPC and gas collection arrangement with a convex array doubling as a cover consistent with embodiments of the present invention. Here, the convex frame **121** functions as the cover **110** with a cover rim **111** that is configured to move towards (and away from) the regolith **112** until it seals against the regolith **112** thereby creating an internal environment **114** that is not in fluid communication with the exterior environment **116** because the regolith **112** is blocking the fluid communication wherein fluid can pass through freely. Accordingly, many of the other attributes in the previous embodiments can be superimposed or otherwise implemented in the embodiment of FIG. 3E without departing from the scope and spirit of the present invention.

(31) FIG. 4A, in view of FIG. 4B, is a line drawing of an isometric view of a CPC **120** and fiberoptic cable **124** arrangement consistent with embodiments of the present invention. As shown, in this configuration, the CPC **120** comprises a concave reflective bowl **130** that is defined between a bowl rim **134** and a bowl apex **136**. The concave reflective bowl **130** is configured to receive sunlight through a CPC inlet aperture **138** defined by the bowl rim **134**, wherein sunlight **140** is reflected to the bowl apex **136** at the bowl apex **136** via the reflective parabolically shaped inner surface **132**. The parabolically shaped inner surface **132** is defined by a 2-D parabola that is rotated in 3-D along the parabola base, or vertex. Certain embodiments envision the concave reflective bowl **130** comprising a reflective inner surface **132** with a refractive index of at least 1.4, which in some embodiments is a mirror. Here, the bowl interior **135** is parabolically shaped to best direct (reflect) the sunlight in a concentrated form to the bowl apex **136** where a light concentrator lens **129** further concentrates the light, which is transmitted to the fiberoptic inlet end **128**. The fiberoptic inlet end **128** comprises a fiberoptic inlet end **128** (also referred to as a light pipe aperture) that receives the concentrated sunlight **143a** and transmits down the length of the fiberoptic cable **124**. The fiberoptic cable **124** channels concentrated light **143a** to a fiberoptic outlet end **126** where the concentrated light **143a** can be emitted as a further concentrated beam of light **143b**.

(32) FIG. 4B is a line drawing of a CPC cross-section depicting solar energy concentration consistent with embodiments of the present invention. The cross-section of the CPC **120** depicts the solar energy **142** from the sun **140** being received through the collector opening/orifice **138** and reflected off the reflective sides **132** to the light concentrator lens **129** at the bowl apex **136**. An aperture in the bowl apex **137** directs the solar energy **142** reflected from the bowl inner sides **132**

to the light concentrator lens **129**, which further concentrates the light **143a** to the fiberoptic inlet end **128**. The acceptance angle 20° **146** of the concave reflective bowl **130** is defined by the diameter of the bowl rim **134**. This determines how much solar energy **140** can be collected, as depicted by the limits of light diameter points **142a** and **142b**. The parabolic shape of the concave reflective bowl **130** is curved to direct and concentrate essentially all the solar energy **142** to the light concentrator lens **129**, as depicted by the reflected angle of the sun's light **142** bouncing off the reflective inner surface **132**. As mentioned, the light concentrator lens **129** focuses the light received from the bowl apex **136** into a further concentrated light **143a** that is channeled into the fiberoptic inlet end **128**. The concentrated light **143a** bounces off the inner surface of the fiberoptic cable **124** through the fiberoptic outlet end **126** where it is emitted. In this embodiment, there is a secondary concentration lens **133** at the fiberoptic outlet end **126**, which focuses the concentrated light **143a** into a high heat light concentrated beam **143b** having a focal point on the regolith **112**. The secondary concentration lens **133** can include a collimating lens, or lens portion, at the fiberoptic outlet end **126**, which causes the emitted light **143b** to be parallel followed by a focusing lens, which causes the parallel light to converge on a focal point. The regolith **112** is heated by the high heat light concentrated beam **143b**, which releases gaseous elements and molecules (gaseous material) from the regolith **112** in the heated region **144**, as shown. The released or otherwise liberated gaseous material migrates into the shielded environment **114** where it is condensed on the gas condensation plates **115** for harvesting.

(33) FIG. **4C** is a line drawing of a compound parabolic concentrator array cross-section in a convex arrangement **100** consistent with embodiments of the present invention. As shown, there are a plurality of CPCs **120** that are distributed along a convex arch, wherein three of the CPCs **120** are positioned to receive sunlight **142** from the sun **140** via their respective CPC inlet apertures **138**. The left two CPCs **120** are not positioned to receive sunlight **142** from the sun **140** based on the angle of the sun **140**. The three CPCs **120** to the right capture and direct the sunlight **140** to the bowl apex **136** where the sunlight **142** is concentrated via the light concentrator lens **129**. The concentrated light **143a** is piped down to the fiberoptic outlet end **126** where it is emitted as a high heat light concentrated beam **143b** at the three far right fiberoptic outlet ends **126**. The high heat light concentrated beam **143b** heats a region **144** of the regolith **112** at sufficient heat to release the gaseous materials trapped in the regolith **112**. In some embodiments, the region **144** is heated to a temperature that is between 600°C .- 1000°C .

(34) Certain embodiments envision the fiberoptic outlet ends **126** being arranged in a line or several lines to heat the regolith **112** in a select region under the high heat light concentrated beams **143b**. One method envisions the rover **102** being stationary with the cover rim **111** on the surface **112a** for enough time for the gas to be liberated from the regolith **112** and collected on the gas condensation plates **115**. The cover rim **111** is then retracted **119** allowing the wagon **102b** to be pulled to a new location to mine fresh regolith **112**. An optional embodiment envisions the fiberoptic outlet ends **126** being actuated by a motor (not shown) to sweep over a larger area of the regolith **112** reducing the amount of raising and lowering the cover rim **111**. Another embodiment envisions the cover rim **111** being a sled that does not need to be raised and lowered, wherein the rover **102** is in constant forward motion with the high heat light concentrated beams **143b** sweeping continuously over the regolith **112**.

(35) Certain embodiments envision a method wherein a CPC and gas collection arrangement comprises an array of CPCs **120** arranged to confront the sun **140** in a manner that essentially permits all or the majority of the CPCs **120** to collect sunlight **142**. For example, consider the CPC and gas collection arrangement **150** of FIG. **1B**, which comprises a half semi-spherical convex CPC array **122b** disposed on a wagon **102b**. Due to its mobility, the wagon **102b** can be positioned with the CPCs **120** essentially optimally pointing towards the sun **140**. In one method embodiment, shown in FIG. **1B**, the CPC and gas collection arrangement **150** is pointed in the east towards the rising sun **140** with the rover **102** traversing the moon **141** in western direction, thereby harvesting

gases from the regolith **112** as the wagon **102b** sweeps across the surface **112a**. The CPC and gas collection arrangement **150** is then turned around with the CPC array **122b** pointing in the west towards the setting sun **140** with the rover **102** traversing the moon **141** in an eastern direction. Because the sun **140** takes approximately 2 weeks to traverse the lunar sky, the CPC and gas collection arrangement **150** can be employed to mine the moon **141** in one direction for the first week and the other direction for the second week of a solar-lunar cycle.

(36) FIGS. 5A and 5B are line drawings illustratively depicting a redirecting spring and nitinol deflector **180** consistent with embodiments of the present invention. As shown, at least one of the plurality of CPCs **120** is connected to a redirecting spring and nitinol deflector **180** that is configured to move at least one CPC **120** towards the sun **140** as it traverses above the wagon **102b**. The outside of the concave reflective bowl **130** is connected to the frame **121** via a nitinol coil **182** on the left side of the bowl **130** and a standard spring **184** (such as a steel spring or titanium spring, etc.) on the right side of the bowl **130**. A voltage source **185** is electrically connected to the ends of the nitinol coil **182** via electrical transmission lines **186**. Nitinol is a shape memory alloy wherein a nitinol structure, such as the nitinol coil **182**, is formed or otherwise set into the compressed coil shape at a temperature around 500° C. In a relaxed state, the coil may be stretched with little resistance but when reheated to its transition temperature, which can be as low as 40° C., the nitinol coil **182** will return to its set coil shape.

(37) As shown in FIG. 5A, the nitinol coil **182** is in a relaxed state wherein the spring **184** comprises enough force to pull the CPC **120** in a first position, which in this embodiment is when the CPC inlet aperture **138** is facing upwards. While in the relaxed state, the voltage source **185** is not providing any current to the nitinol coil **182**.

(38) As shown in FIG. 5B, the nitinol coil **182** is compressed in its memory set state, which is accomplished by heats the nitinol coil **182** above its transition temperature with current supplied to the ends of the nitinol coil **182** via the electrical transmission lines **186**. The heated nitinol coil **182** is stronger than (i.e., overcomes) the spring force of the spring **184** thereby pivoting the CPC **120** at the bowl apex **136** into a second position tipping the CPC inlet aperture **138** to the left. In this way, the CPC can be made to better confront the sun **140** thereby improving the solar energy collection to generate heat from the sunlight transmission down the fiberoptic cable **124**.

(39) The redirecting spring and nitinol deflector **180** is one embodiment of a CPC deflector arrangement, which could be accomplished by a hydraulic piston arrangement or motor and linkage assembly to move the CPC **120** to track the sun **140**. One advantage that the redirecting spring and nitinol deflector **180** has over these other optional embodiments is that the nitinol and spring have no friction interfacing surfaces, which might wear due to the abrasive nature of regolith dust.

(40) Another embodiment presented herein that can be used with the CPC arrangement, or optionally be used without the CPC arrangement as a stand-alone system, is a heat recovery arrangement for collecting vaporized gas trapped in regolith. The heat recovery arrangement generally comprises a rover that carries heat recovery elements that cooperate with a primary heat source. The heat recovery elements include a preheat contact element that preheats a region of regolith before the region is brought to high heat configured to liberate target gases in the regolith by the primary heat source. As the rover moves forward, the preheat contact element receives heat collected from the high heat region via a heat recovery sled that moves in contact with the high heat region. Heat is transferred between the heat recovery sled and the preheat contact element via a heat transfer medium that circulates through the heat recovery sled and preheat contact element.

(41) FIGS. 6A-6G are line drawings illustratively depicting a heat recovery arrangement **240** that can be used in conjunction with the CPC and gas collection arrangement **100** consistent with embodiments of the present invention. FIG. 6A illustratively depicts a block diagram of an isometric line drawing of a rover **102** showing the six rover wheels **208**, a heating element port **109** in the rover top **107** and the rover front end **108A** and back end **108B**. As shown by arrow **242**, the rover **102** moves in a (forward) direction leading with the front end **108A** and trailing with the rear

end **108B**. Described in more detail below are heat recovery elements **250** located under the rover **102**, i.e., located at the rover underside **244**.

(42) FIG. **6B** is a side view cross-section line drawing of the heat recovery arrangement **240** consistent with embodiments of the present invention. The rover **102** is shown as a block diagram configuration for simplification. For example, the core portion **238** of the rover **102** is depicted as being a solid block between the top surface **107** and the underneath side **244** with a heat element port **109** passing therethrough. In actuality, however, the rover cover **102** will likely be sheet metal, carbon fiber, or some other suitable light weight material shell which includes other elements (not shown) taking up the space, such as cryogenic gas collection surfaces equipped to collect gas captured in the gas collecting shield **252**. The gas collecting shield **252** is configured to either rest on the regolith **112**, or some other granular soil surface, or be in close proximity to the regolith surface, such as within a fraction of an inch. For reference, a rover wheel **208** is shown extending from the rover rear end (trailing edge) **108B** in addition to the wheel hubs and axle system **208B** shown under the core portion **238**. In operation, the heat recovery arrangement **240** takes advantage of heat collection when the rover **102** is moving in a forward direction as indicated by the arrow **242**.

(43) FIG. **6C** is a block diagram of the heat recovery/exchange arrangement **240** described in view of FIG. **6B**. The heat recovery elements **250** are configured to preheat a region of regolith **144** before that region of regolith **144** is brought up to a target temperature via a primary heating source **245**. The target temperature being high enough to liberate target gases, such as He-3, trapped in the regolith **112**. By preheating the region **144** prior to using the primary heating source **245**, the amount of energy needed to bring the region **144** up to the target temperature is reduced, which conserves energy in the system **240**. In other words, the change in temperature between starting temperature and target temperature (in the gap **264**) is lowered thereby lowering the energy required to liberate the target gases from the regolith **112**.

(44) The heat recovery elements **250** are arranged to recover heat from the heated region **144** and reuse the recovered heat to preheat a new region that will soon be subjected to heat from the primary heating source **245**. Hence, the primary heat source **245**, that extends through the heat element port **109**, heats a region **144** in the regolith **112** in a gap **264** (defined between a heat recovery sled **280** and a preheat contact element **260**) to a target temperature, such as 850° C. or some other temperature that may be between 600° C.-1000° C., for example. Certain embodiments envision the primary heat source **245** being the concentrated heat light beam **143b**, such as that shown in FIG. **4B**, or some other heat element such as a laser, ultrasonic heat element, radiant heater, contact heat element, etc. The primary heat source **245** heats a region **144** of the regolith **112** at sufficient temperature to release the gaseous materials trapped in the regolith **112**.

(45) The heat recovery elements **250** include the heat recovery sled **280** separated from the preheat contact element **260** by a primary heating gap **264**, which could be several inches (e.g., 6 inches) to several feet (e.g., 3 feet) across. The preheat contact element **260**, located at the underside **244** near the rover leading edge (front end) **108A**, is configured to contact and preheat the regolith **112**. Certain embodiments envision the preheat contact element **260** and heat recovery sled **280** extending beyond the perimeter of the rover **102**. In this embodiment, the preheat contact element **260** is a rake comprising a plurality of bladed plates **266** that dig into the regolith **112** to provide both conductive and radiative heat transfer to the regolith **112**. The heat recovery sled **280**, located on the trailing **108B** underside **244** of the rover **102**, is also configured to contact the regolith **112** and pull residual heat left over from the heat produced by the primary heat source **245**. In this embodiment, the heat recovery sled **280** comprises a plurality of rails **286** (see FIG. **6E**) each with a rail leading edge **282** configured to dig into the regolith **112**. The sled rails **286** increases the surface area of where the heat recovery sled **280** interfaces the regolith **112** for more efficient heat collection via radiation and conduction. The collected heat is transferred to the preheat contact element **260** via a liquid heat transfer medium **305** that cycles through the heat recovery sled **280**

via a heat exchange tube arrangement **300**. A gas collecting shield **252** covers at least the heating gap **264**, and in this embodiment, at least a part of the preheat contact element **260**. The gas collecting shield **252** is configured to contain gas liberated from the regolith **112** heated by the primary heat source **245**.

(46) As shown in connection with FIGS. **6D1-6D3** in view of FIG. **6B**, the primary heat source **245** heats a first region **144A** of regolith **112** to a temperature of 850°C . (or some other suitable temperature that will release target gases from the regolith **112**) as shown in FIG. **6D1**. As the rover **102** moves forward **242**, the heat recovery sled **280** moves into contact with the first heated region **144A**, which may have cooled down a little bit by the time it reaches the heat recovery sled **280** but will still be reasonably high. In this example, the heated region cools to 825°C ., as shown in the example of FIG. **6D2**. Meanwhile, the primary heat source **245** heats a second region **144B** to a temperature of 850°C . As the rover **102** continues to move forward **242**, as shown in FIG. **6D3**, heat from the first region **144A** heats up the heat recovery sled **280** via conduction and radiative heat transfer while the second region **144B** moves in contact with the heat recovery sled **280**. The heat recovered from the first region **144A** by the heat recovery sled **280** is transferred to the preheat contact element **260** via a liquid heat transfer medium **305** cycled through the heat recovery sled **280** and the preheat contact element **260**. The heated medium **305A**, of FIG. **6F**, exchanges its heat to the preheat contact element **260**, i.e., the heated medium **305A** heats up the preheat contact element **260**, which in turn preheats a third region **144C** of the regolith **112**. As further shown in FIG. **6D3**, the primary heat source **245** continues to heat the gap region **264**, which is depicted by the heated region **144B'**. As the rover **102** continues to move forward **242**, naturally, the third region **144C** will move in front of the primary heat source **245** already at an elevated temperature (in this example a temperature of 500°C .), thereby reducing the amount of energy required to bring the third region **144C** up to the target temperature. Some embodiments envision the rover **102** moving continuously while other embodiments envision the rover **102** stopping and starting (stuttered forward motion).

(47) FIG. **6E** is an isometric cross-section view of the heat recovery/exchange arrangement **240** shown in FIG. **6B** consistent with embodiments of the present invention. The angled view of FIG. **6E** provides a different perspective of the heat recovery elements **250** and gas collecting shield **252**, which is partially sectioned to provide a better view of the heat recovery elements **250**. This perspective shows the preheat plates teeth **262**, which slice into the regolith **112** as the rover **102** moves forward. The preheat plates (rake) **266** also increase the surface area of the preheat contact element **260** to increase heat exchange. The heat recovery sled **280** comprises a plurality of rails **286** each with a rail leading edge **282** that digs/slides into the regolith **112**. Like the preheat plates **266**, the sled rails **286**, when embedded in the regolith **112**, increases the surface area of where the sled **280** interfaces the regolith **112**, which produces more efficient heat collection via radiation and conduction. As mentioned earlier, the cover rim **111** can be at or within an inch or two of the regolith **112**.

(48) FIG. **6F** is an isometric view of the heat recovery elements **250** showing the feed line **304** and the return line **302** cross sectioned. In this embodiment, the feed line **304** and the return line **302** link the heat recovery sled **280** with the preheat contact element **260**, which are separated by a heating gap **264**. The primary heat source heated region **144** is shown via a shaded ellipse in the heating gap **264**. As shown here, there are two heat exchange tubes **300**, a first heat exchange tube **300A** on the left side of the heating elements **260/280** and a second heat exchange tube **300B** on the right side of the heating elements **260/280**. Each heat exchange tube **300** is a closed system that contains a heat exchange medium **305**. Non limiting examples of a heat exchange medium (working fluid) **305** includes the use of liquid metals (metals that are in a liquid form above their melting point), such as mercury, cesium, rubidium, potassium, sodium, lithium, calcium, lead, indium, silver, for example. A publication incorporated herein by reference entitled "A review of liquid metal high temperature heat pipes: Theoretical model, design, and application" by Tian et al.,

published in the International Journal of Heat and Mass Transfer provides examples and thermodynamic analysis of heat pipes using liquid and vapor metals. Table 1 below is a listing of various liquid metals **305** along with their relative melting points, boiling points and operation temperature range. It should be noted that the metal when in essentially a solid/non-liquid state can be heated by an auxiliary electric heater (not shown) to melt the frozen liquid metal.

(49) TABLE-US-00001 TABLE 1 Working Melting Point Boiling Point Operation Fluid at 1 atm (K) at 1 atm (K) Temperature Range Mercury (Hg) 234.2 630.1 523-923 Cesium (Cs) 301.6 943.0 723-1173 Rubidium (Rb) 312.7 959.2 800-1275 Potassium (K) 336.4 1032 773-1273 Sodium (Na) 371.0 1151 873-1473 Lithium (Li) 453.7 1615 1273-2073 Calcium (Ca) 1112 1762 1400-2100 Lead (Pb) 600.6 2013 1670-2200 Indium (In) 429.7 2353 2000-3000 Silver (Ag) 1234 2485 2073-2573

(50) With continued reference to FIG. **6F** and with respect to the first heat exchange tube **300A**, a heat collecting coil **306** is embedded in the heat recovery sled **280** and a heat emitting coil **308** is embedded in the preheat contact element **260**. The heat collecting coil **306** loops between the heat emitting coil **308** via a feed line **304** and a return line **302**. The feed line **304** connects to the heat collecting coil **306** at a first junction **291** at the sled trailing edge **298**. The feed line **304** further connects to the heat emitting coil **308** at a fourth junction **297** at the preheat contact element leading edge **292**. The return line **302** connects the heat collecting coil **306** at a second junction **293** at the sled leading edge **296** to the heat emitting coil **308** at a third junction **295** at the preheat contact element trailing edge **294**. In operation, the heat recovery sled **208** picks up heat from the heated region **144** of the regolith **112** and heats the cooled medium **305B** received from the preheat contact element **260** via the feed line **304**. The cooled medium **305B** is heated as flows through the heat collecting coil **306** towards the return line **302**. The heated medium **305A** is essentially its maximum temperature upon exiting the heat collecting coil **306** at the sled leading edge **296**. The heated medium **305A** flows into the preheat contact element **260** at the third junction **295** via the return line **302**. Heat is exchanged to the preheat contact element **260** as the heated medium **305A** flows through the heat emitting coil **308** to the preheat contact element leading edge **292**. Heat is exchanged from the preheat contact element **260** to the regolith **112**, thereby cooling down the heat transfer medium **305** in the heat emitting coil **308**. The heated preheat contact element **260** (heated by the heated medium **305A**), in turn, exchanges or otherwise heats up the regolith **112** through conduction and radiation. The heat transfer medium **305** is cooled to essentially its minimum temperature upon exiting the heat emitting coil **308** at the preheat contact element leading edge **392**. The cooled medium **305B** is returned to the heat recovery sled **280** via the return line **302**. Certain embodiments contemplate the heat transfer medium **305** being a liquid flowing through a wicking structure **303**, such as capillary tubes shown in FIG. **6G**, in the return line **302** but a vapor in the feed line **304**. FIG. **6G** is a closeup of a portion of the return line **302** and feed line **302** of the first heat exchange tube **300A** at the heat recovery sled **280**.

(51) FIG. **6H** is an active circulation system connected to the heat exchange tubes consistent with embodiments of the present invention. Unlike the passive wicking elements **303** shown in FIGS. **6F** and **6G**, FIG. **6H** depicts an active pump **270** in line with the first and second heat exchange tubes **300A** and **300B**. Though the pumps **270** can reside in other locations along the heat exchange tubes **300A** and **300B**, in this embodiment, the pumps **270** are in a preheat housing **268** at the front of the preheat contact element **260** just above the preheat plates **266**, as shown. In this embodiment the pumps **270** are connected to their respective return lines **302**. The heat transfer medium **305** is circulated by the pumps **270** from the heat emitting coil **308** through the heat emitting coil exit port **272** and into the return line **302**. It should be appreciated that with a pump **270** there will be no wicking element **303**, rather the pumping/active system works with an unobstructed return tube.

(52) With the present description in mind, below are some examples of certain embodiments illustratively complementing some of the apparatus embodiments discussed above and presented in the figures to aid the reader. Accordingly, the elements called out below are provided by example to

aid in the understanding of the present invention and should not be considered limiting. The reader will appreciate that the below elements and configurations can be interchangeable within the scope and spirit of the present invention. The illustrative embodiments can include elements from the figures.

(53) In that light, certain embodiments of the present invention envision a heat recovery arrangement **240** generally comprising a rover **102** carrying a primary heat source **245** that cooperates or otherwise works/functions with an arrangement of heat recovery elements **250**. The rover **102** can be defined as having wheels or tracks **208** that moved the rover **102** over an extra-terrestrial surface. The rover can be defined between a rover front end **108A** and a rover rear end **108B**, wherein the rover **102** is configured to travel in a direction **242** leading with the rover front end **108A**. The heat recovery elements **250** include a preheat contact element **260** and a heat recovery sled **280**, both of which are configured to rest atop granular soil **112**. The heat recovery elements **250** further include a gap **264** spacing apart the preheat contact element **260** from the heat recovery sled **280**, wherein the preheat contact element **260** is closer to the rover front end **108A** than the heat recovery sled **280**. The heat recovery arrangement **240** further includes a primary heat source **245** (configured to heat the granular soil **112** in the gap **264**) and a gas collecting shield **252** covering at least the gap **264**. The heat recovery elements **250** further include a heat exchange tube **300** comprising a feed line **304**, a return line **302**, a heat collecting coil **306**, and a heat emitting coil **308**, wherein the feed line **304** is configured to flow heated medium **305A** from a heat collecting coil **306** to the heat emitting coil **308**, the return line **302** is configured to flow cooled medium **305B** from the heat emitting coil **308** to the heat collecting coil **306**. The heat recovery sled **280** contains the heat collecting coil **306** and the preheat contact element **260** contains the heat emitting coil **308**. The gas collecting shield **252**, the preheat contact element **260** and the heat recovery sled **280** are located under the rover **102**.

(54) The heat recovery arrangement **240** further envisions the gas collecting shield **252** covering at least part of the contact heat source **260** while other embodiments envision the gas collecting shield **252** basically encapsulating the space between the regolith **112** in the gap **264**.

(55) In some embodiments, the heat recovery arrangement **240** envisions the primary heat source **245** being configured to heat the granular soil in the gap **264** via a light beam **143b**.

(56) The heat recovery arrangement **240** further imagines the primary heat source **245** being connected to at least one solar collector **120**, which provides the energy that is converted to heat.

(57) The heat recovery arrangement **240** further contemplates the preheat contact element **260** comprising a plurality of blades **266** that are configured to penetrate the granular soil **112** linearly in the direction **242**.

(58) In one embodiment, the heat recovery arrangement **240** envisions the heat collecting coil **306** being embedded in the heat recovery sled **280** and the heat emitting coil **308** being embedded in the preheat contact element **260**.

(59) Another embodiment contemplates the heat exchange tube **300** of the heat recovery arrangement **240** further comprising a pump **270** that is configured to circulate the heated medium **305A** and the cooled medium **305B**.

(60) The heat recovery arrangement **240** can optionally use a liquid metal as the medium **305** for heat transfer.

(61) In other embodiments, the heat recovery arrangement **240** imagines, when in operation, the heated medium **305A** being in a gas phase when and the cooled medium **305B** being in a liquid phase. The heated medium **305A** is defined as the medium **305** when transitioning to the preheat contact element **260** via the feed line **304** and the cooled medium **305B** is defined as the medium **305** when transitioning to the heat recovery sled **280** via the return line **302**.

(62) The heat recovery arrangement **240** further contemplates the return line **302** comprising a plurality of capillary tubes **303**, porous wicking structures that use capillary action but are not necessarily tubes.

(63) The heat recovery arrangement **240** is envisioned having the preheat contact element **260** providing heat to the granular soil **112** before the primary heat source **245** heats the granular soil **112** in the gap **264**. The granular soil **112** can be regolith in lunar applications.

(64) Still other embodiments envision a heat recovery system **240** generally comprising a rover **102** carrying a primary heat source **245** that cooperates with an arrangement of heat recovery elements **250**, the rover **102** being configured to move in a forward direction **242**. The heat recovery elements **250** can include a preheat contact element **260** that is configured to contact and preheat granular soil **112** and a heating gap **264** between the preheat contact element **260** and a heat recovery sled **280**. The heat recovery system **240** further includes a primary heat source **245** that is configured and arranged to heat a region **144** of the preheated granular soil **112B** in the gap **264**. The heat recovery sled **280** is configured and arranged to collect at least a portion of the heat from the region **144** when the rover **102** moves the heat recovery sled **280** in contact with the region **144**. The heat recovery system **240** further includes a gas collecting shield **252** covering at least the gap **264**, wherein the gas collecting shield **252** is configured to contain gas liberated from the granular soil **112** when the granular soil **112** is heated by the primary heat source **245**. The heat recovery elements **250** further include a heat exchange tube **300** comprising a feed line **304** that is configured to carry warmed medium **305A** from a heat collecting coil **306** in the preheat contact element **260** to a heat emitting coil **308** in the heat recovery sled **280**, and a return line **302** that is configured to carry cooled medium **305B** from the heat emitting coil **308** to the to the heat collecting coil **306**. The gas collecting shield **252**, the preheat contact element **260** and the heat recovery sled **280** are located on an under side **244** the rover **102**.

(65) The heat recovery system **240** further envisions the preheat contact element **260** being configured to preheat the region **144** to a first temperature and wherein the primary heat source **245** is configured to heat the region of granular soil **144** to a second temperature that is higher than the first temperature when the region of granular soil **144** moved to the gap **264**.

(66) The heat recovery system **240** further imagines the preheat contact element **260** and the heat recovery sled **280** being configured to contact the granular soil **112**.

(67) The gas collecting shield **252** in the heat recovery system **240** can further cover at least part of the contact heat source **260**.

(68) The heat recovery system **240** further contemplates the preheat contact element **260** comprising a plurality of preheat plates **266** that are configured to penetrate and drag through the granular soil **112** along the forward direction **242** (in essentially a linear direction not to be confused with a spinning or drilling bit).

(69) Another embodiment of the present invention contemplates a method for using rover **102** supporting a heat recovery system **240** wherein the method generally comprises steps for heating granular soil **112**, such as regolith, while moving the rover **102**. More specifically, the method can include a heating step for heating a first region of granular soil **144A** in a gap **264** between a preheat contact element **260** and a heat recovery sled **280** with a primary heat source **245**. Moving the heat recovery sled **280** in contact with the first region of granular soil **144A**. Warming a medium **305A** that is circulating through a heat collecting coil **306** while the heat recovery sled **280** is in contact with the first region of granular soil **144A**. The coil being embedded in or otherwise snaking through the the heat recovery sled **280**. The warming step is accomplished by collecting a portion of the heat from the heated granular soil **112B**. The method further includes preheating a second region of granular soil **144B** by circulating the warmed medium **305A** through a heat emitting coil **308** embedded in the preheat contact element **260**. Moving the gap **264** over the preheated second region of granular soil **144B** and heating the preheated second region of granular soil **144B** via the primary heat source **245**.

(70) The method can further comprise a step for containing gas liberated from the heated first region of granular soil **144A** in a gas collecting shield **252** that covers the gap **264**. Some embodiments contemplate the gas collecting shield **252**, the preheat contact element **260** and the

heat recovery sled **280** being located under the rover **102**.

(71) The method further envisions the primary light source **245** being a beam of light **143b**.

(72) Still other embodiments of the present invention envision a CPC heating arrangement **100** that generally comprises a transporter **102a** that carries a plurality of CPCs **120** that are equipped to heat granular soil **112** to mine target gases trapped in the granular soil **112**, as depicted largely in FIG. 3A. More specifically, the transporter **102a** comprises a transporter body **109** defining a top end **107** and a bottom end **101**, wherein the bottom end **101** is configured to interface a surface **112A** of an extra-terrestrial body **141**. The transporter **102a** is configured to be moved in different locations on the extra-terrestrial body **190**, such as the moon **141**. The CPC arrangement further comprises a cover **110** having an internal cover volume **114**, which is a volume within the cover **110** defined from between a cover rim **111** and an inside cover top **107c** of the cover **110**. The internal cover volume **114** is defined by a shielded environment **114** when the cover rim **111** rests atop granular soil **112**. The shielded environment **114** is the internal volume of the cover and is only in communication with an outside environment **116** via the cover rim **111** when the cover rim **111** is not resting atop the granular soil **112**. Resting atop means in contact with the granular soil wherein there are essentially no gaps between the granular soil surface **112a** and the cover rim **111**. Resting can be wherein at least a portion of the weight of the cover **110** is supported by the granular soil **112** or if the cover **110** is pressed into the granular soil **112** via a motorized actuator connected to the transporter **102a**. The plurality of CPCs **120** extends from the top end **107**, such as an array **122** on an armature or frame **121**. Each CPC **120** comprises a concave reflective bowl **130** having a refractive index of at least 1.4. Each of the concave reflective bowls **130** are defined between a bowl rim **134** and a bowl apex **136**, wherein the bowl rim **134** is configured to confront the sun **140** and the bowl apex **136** (an aperture located at the bowl apex **136**) is configured to receive sunlight **142** from the concave reflective bowl **134**. Each of the concave reflective bowls **130** further comprises a fiberoptic cable **124** extending from a fiberoptic inlet end **128** connected to the bowl apex **136** to a fiberoptic outlet end **126** located in the internal volume **114**. The fiberoptic outlet end **126** is configured to reside within 6 inches from the granular soil **112**. The sunlight **142** is configured to be received at the fiberoptic inlet end **128** and emitted at the fiberoptic outlet end **126**.

(73) Another embodiment of the present invention contemplates a CPC arrangement **100** generally comprising a transporter **102a**, an internal cover volume **114** and a cryogenically cooled condensation surface **115**. The transporter **102a** supports a plurality of CPCs **120** each configured to receive sunlight. Each CPC **120** comprises a concave reflective bowl **130** defined between a bowl rim **134** and a bowl apex **136** comprising a bowl aperture and a fiberoptic cable extending from a fiberoptic inlet end **128** at the apex **136** to a fiberoptic outlet end **126**. The fiberoptic outlet end **126** is configured to emit the sunlight, that is received at the fiberoptic inlet end **128** from the bowl aperture, on a granular surface **112**. The CPC arrangement **100** further comprises an internal cover volume **114** defined within a cover **110** that extends from an internal cover top **107c** to a cover rim **111**. There is a cryogenically cooled surface **115** within the internal cover volume **114**, wherein the cryogenically cooled surface **115** configured to be cooled by cryogenic liquid. The fiberoptic outlet end **126** extends into the internal cover volume **114**.

(74) The CPC arrangement **100** further envisions the cover **110** being connected to the transporter **102a**.

(75) The CPC arrangement **100** further imagines the transporter **102a** being a rover **102**.

(76) The CPC arrangement **100** further contemplates the plurality of CPCs **120** being arranged in a semi-sphere on at least one side of the transporter **102a**.

(77) The CPC arrangement **100** further considers the plurality of CPCs **120** being arranged in a convex array **122** on at least one side of the transporter **102a**. Here, the convex array **122** can optionally be bowl-shaped **162**.

(78) The CPC arrangement **100** further envisions each of the concave reflective bowls **130** being

parabolic-shaped.

(79) The CPC arrangement **100** contemplates at least one of the plurality of concave reflective bowls **130** being connected to a redirector, such as a redirecting spring and nitinol deflector arrangement **180**, that is configured to move the at least one concave reflective bowls **130** to better confront the sun **140** as it traverses above the transporter **102a**. The redirector can be a nitinol linkage that moves when heated and cooled, at least one hydraulic lifter, a motor driven linkage that moves each CPC **120**, etc.

(80) The CPC arrangement **100** further envisions an embodiment wherein the plurality of concave reflective bowls **130** are statically mounted on the transporter **102a**, meaning they cannot be adjusted or moved to track the sun **140**, for example.

(81) The CPC arrangement **100** further envisions the transporter **102a** being configured to traverse the extra-terrestrial body **141** in a first direction **195** while the sun **140** is in a rising trajectory and the transporter **102a** being configured to traverse the extra-terrestrial body **141** in a second direction **196** while the sun **140** is in a setting trajectory.

(82) The CPC arrangement **100** further imagines at least one of the fiberoptic cables **124** being connected to a fiberoptic redirector or motor that is configured to move or bend the at least one fiberoptic cable **124** to shine the sunlight **142** over a region **144** of the granular soil **112**.

(83) The CPC arrangement **100** further contemplates the fiberoptic outlet ends **126** being arranged in a line that is configured to traverse the granular soil **112** (like the teeth of a rake) as the transporter **102a** traverses the extra-terrestrial body **141**.

(84) The CPC arrangement **100** can further comprise a light concentrator lens **129** configured to receive the sunlight **142** at the bowl apex **136** and concentrate the sunlight **142** into concentrated light **143a** that is transmitted through the fiberoptic cable **124**.

(85) The CPC arrangement **100** can further comprise a secondary concentration lens **133** that is configured to receive the sunlight **142** from the fiberoptic outlet end **126** and concentrate the sunlight **142** into a high heat light concentrated beam **143b** that is directed to the granular soil **112**.

(86) Yet another embodiment of the present invention contemplates a CPC system **100** that generally comprises a transporter **102a** that carries a plurality of CPCs **120** that are equipped to heat regolith **112** to mine target gases trapped in the regolith **112**. More specifically, the transporter **102a** defines a top end **107** and a bottom end **101** with the bottom end being configured to interface regolith **112**. The transporter **102a** configured to be moved in different locations on the moon **141**. The CPC system **100** further comprises a cover **110** that defines a shielded environment **114** when a cover rim **111** of the cover **110** rests atop the regolith **112**. The shielded environment **114** (or the internal cover volume when not in contact with the regolith **112**) is only in communication with an outside environment **116** via the cover rim **111** when the cover **110** does not rest atop the regolith **112**. The plurality of CPCs **120** extends from the top end **107**, wherein each CPC **120** comprises a concave parabolic reflective bowl **130**. Each of the concave reflective bowls **130** is defined between a bowl rim **134** and a bowl apex **136**, wherein the bowl rim **134** is configured to receive sunlight. An aperture in the bowl apex **136** is configured to receive the sunlight **142** from the concave reflective bowl **134**. A fiberoptic cable **124** extends from a fiberoptic inlet end **128** to a fiberoptic outlet end **126** and connects to the bowl apex **136** at the fiberoptic inlet end **128**. The fiberoptic cable **124** is in light communication with the aperture, meaning light moves through the aperture and into the fiberoptic cable **124**. The fiberoptic outlet end **126** is located in the shielded environment **114** and is configured to interface the regolith **112**. The fiberoptic cable **124** is configured to transmit the sunlight **142** from the fiberoptic inlet end **128** through the fiberoptic outlet end **126**.

(87) The CPC system **100** further envisions the shielded environment **114** extending from a wagon **102b** that is configured to be moved by a rover **102**.

(88) The CPC system **100** further imagines the transporter **102a** being a rover **102** or a wagon **102b**.

(89) The CPC system **100** can further comprise a light concentrator lens **129** that bridges the aperture and the fiberoptic inlet end **128**.

(90) The CPC system **100** can further comprise a secondary concentration lens **133** that is configured to focus light from the fiberoptic outlet end **126**.

(91) Still, another embodiment of the present invention contemplates a CPC and gas collection arrangement **100** that generally comprises a transporter **102a** that carries a plurality of CPCs **122** that are equipped to heat granular soil **112** to mine target gases trapped in the granular soil **112**. More specifically, the transporter **102a** is configured to be moved in different locations on the moon **141**. The arrangement **100** further comprises a cover **110** that defines a shielded environment **114** when a cover rim **111** of the cover **110** rests atop the regolith **112**. The shielded environment **114** is not in communication with an outside environment **116**. The plurality of CPCs **122**, wherein each of the plurality of CPCs **120** comprises a concave parabolic reflective bowl **130** that is defined between a bowl rim **134** and a bowl apex **136**. The bowl rim **134** is configured to receive sunlight and direct the sunlight **142** to an aperture at the bowl apex **136**.

(92) The arrangement **100** can further comprise a fiberoptic cable **124** that has a fiberoptic inlet end **128** extending from the aperture and a fiberoptic outlet end **126** that extends into the shielded environment **114**. The fiberoptic outlet end **126** is configured to interface the regolith **112**, wherein the sunlight **142** is configured to be received at the fiberoptic inlet end **128** and emitted at the fiberoptic outlet end **126**.

(93) It is to be understood that even though numerous characteristics and advantages of various embodiments of the present invention have been set forth in the foregoing description, together with the details of the structure and function of various embodiments of the invention, this disclosure is illustrative only, and changes may be made in detail, especially in matters of structure and arrangement of parts within the principles of the present invention to the full extent indicated by the broad general meaning of the terms in which the appended embodiments are expressed. For example, the orientation of the elements and the plate can include other geometries not explicitly shown in the embodiments above while maintaining essentially the same functionality without departing from the scope and spirit of the present invention. Likewise, the materials and construction of the cooling surfaces can vary but serve the same purpose without departing from the scope and spirit of the present invention. It should further be appreciated that the arrangement of the CPCs **120**, the cover, the transporter, and the arrangement of the fiberoptic cables can vary while maintaining the functionality described within the scope and spirit of the present invention. Moreover, the electronics and computing that enable the functionality of the gas collection system are not described in detail because the electronics and computing elements either exist or are easily constructed by those skilled in the art.

(94) It will be clear that the present invention is well adapted to attain the ends and advantages mentioned as well as those inherent therein. While presently preferred embodiments have been described for purposes of this disclosure, numerous changes may be made which readily suggest themselves to those skilled in the art and which are encompassed in the spirit of the invention disclosed and as defined in the appended claims.

Claims

1. A heat recovery arrangement comprising: a rover defining a rover front end and a rover rear end, wherein the rover is configured to travel in a direction leading with the rover front end; both a preheat contact element and a heat recovery sled configured to rest atop granular soil; a gap spacing apart the preheat contact element from the heat recovery sled, wherein the preheat contact element is closer to the rover front end than the heat recovery sled; a primary heat source configured to heat the granular soil in the gap; a gas collecting shield covering at least the gap; and a heat exchange tube comprising a feed line, a return line, a heat collecting coil, and a heat emitting coil, wherein;

- the feed line is configured to flow heated medium from the heat collecting coil to the heat emitting coil, the return line is configured to flow cooled medium from the heat emitting coil to the heat collecting coil, the heat recovery sled contains the heat collecting coil and the preheat contact element contains the heat emitting coil, and the gas collecting shield, the preheat contact element and the heat recovery sled are located under the rover.
2. The heat recovery arrangement of claim 1, wherein the gas collecting shield further covering at least part of the preheat contact element.
 3. The heat recovery arrangement of claim 1, wherein the primary heat source is configured to heat the granular soil in the gap via a light beam.
 4. The heat recovery arrangement of claim 1, wherein the primary heat source is connected to at least one solar collector.
 5. The heat recovery arrangement of claim 1, wherein the preheat contact element comprises a plurality of preheat plates configured to penetrate the granular soil linearly in the direction of travel.
 6. The heat recovery arrangement of claim 1, wherein the heat collecting coil is embedded in the heat recovery sled and the heat emitting coil is embedded in the preheat contact element.
 7. The heat recovery arrangement of claim 1, wherein the heat exchange tube further comprises a pump configured to circulate the heated medium and the cooled medium.
 8. The heat recovery arrangement of claim 1, wherein the medium is a liquid metal.
 9. The heat recovery arrangement of claim 1, wherein the heated medium is configured to operate in a gas phase and the cooled medium is configured to operate in a liquid phase.
 10. The heat recovery arrangement of claim 1, wherein the return line comprises a plurality of capillary tubes.
 11. The heat recovery arrangement of claim 1, wherein the preheat contact element is configured to provide heat to the granular soil before the primary heat source heats the granular soil in the gap.
 12. A heat recovery system comprising: a rover configured to move in a forward direction; a preheat contact element configured to contact and preheat granular soil; a heating gap between the preheat contact element and a heat recovery sled; a primary heat source configured to heat a region of the preheated granular soil in the gap, the heat recovery sled configured to collect at least a portion of the heat from the region when the rover moves the heat recovery sled in contact with the region; a gas collecting shield covering at least the gap, wherein the gas collecting shield is configured to contain gas liberated from the granular soil when heated by the primary heat source; and a heat exchange tube comprising: a feed line configured to carry warmed medium from a heat collecting coil in the preheat contact element to a heat emitting coil in the heat recovery sled, and a return line configured to carry cooled medium from the heat emitting coil to the heat collecting coil, the gas collecting shield, the preheat contact element and the heat recovery sled are located on an under side of the rover.
 13. The heat recovery system of claim 12, wherein the preheat contact element is configured to preheat the region to a first temperature and wherein the primary heat source is configured to heat the region to a second temperature that is higher than the first temperature when the gap is moved to the region.
 14. The heat recovery system of claim 12, wherein the preheat contact element and the heat recovery sled are configured to contact the granular soil.
 15. The heat recovery system of claim 12, wherein the gas collecting shield further covering at least part of the contact heat source.
 16. The heat recovery system of claim 12, wherein the preheat contact element comprises a plurality of preheat plates configured to penetrate and drag through the granular soil along the forward direction.
 17. A method for using a heat recovery system, the method comprising: heating a first region of granular soil in a gap between a preheat contact element and a heat recovery sled with a primary

heat source; moving the heat recovery sled in contact with the first region of granular soil; warming a medium circulating through a heat collecting coil that is embedded in the heat recovery sled while the heat recovery sled is in contact with the first region of granular soil; preheating a second region of granular soil by circulating the warmed medium through a heat emitting coil embedded in the preheat contact element; and moving the gap over the preheated second region of granular soil and heating the preheated second region of granular soil via the primary heat source.

18. The method of claim 17 further comprising containing gas liberated from the heated first region of granular soil in a gas collecting shield covering the gap.

19. The method of claim 18, wherein the gas collecting shield, the preheat contact element and the heat recovery sled are located under the rover.

20. The method of claim 17, wherein the primary light source is a beam of light.
